

ZK-Shapley Protocol Overview

Retailer
(private D_A)

Distributor
(private D_B)

Manufacturer
(private D_C)

Federated ridge regression: each firm submits
sufficient statistics (A_i, b_i) + commit-reveal binding

Smart contract: verify commitments,
compute Shapley payoffs $\phi_i(v)$

Theorem 1 (Efficiency): $\sum_i \phi_i = v(N)$ | Theorem 2 (Fairness axioms) | structural robustness to size misreport